



Medical Imaging Clinical AI

EfficientNet-B0 chest X-ray classification demo with model evaluation and local inference.

Research and portfolio demonstration only. This application is not for diagnosis or patient care.

Predict Results **Clinical Interpretation** How to Run

Clinical Interpretation

Project Question

Can combining medical images with structured clinical metadata improve disease-risk prediction compared with image-only AI?

Decision Support, Not Diagnosis

This project is a clinical AI decision-support demonstration. It is **not** a diagnostic medical device and should not be used to make real clinical decisions.

What Metadata Improves

Structured clinical metadata can provide context that is not fully visible in the medical image. Examples include age, sex, prior conditions, acquisition site, clinical history, and other structured patient-level features.

In this demo, MedMNIST provides image-label data. Because MedMNIST does not provide full clinical metadata, the package creates synthetic metadata for demonstration. The code is designed so real metadata can later replace the synthetic metadata CSV.

Where the Model Can Fail

The model may fail when training data is small, labels are noisy, imaging conditions differ from training data, metadata is missing or biased, or the model learns shortcuts unrelated to disease.

Fairness and Bias Concerns

Clinical AI can perform differently across demographic groups, imaging sites, and patient populations. Metadata may improve prediction but may also introduce bias if sensitive variables or proxies are used without careful validation.

Fairness checks should include subgroup performance by age, sex, scanner site, and clinical group.

Need for Clinician Review

A real clinical deployment would require clinician review, external validation, calibration analysis, fairness testing, privacy/security review, and regulatory review.

Summary

The multimodal model is useful as a portfolio demonstration because it reflects a more realistic clinical workflow than image-only classification. Predictions should always be interpreted with caution and reviewed by qualified clinical experts.